L1a: Course Orientation, Markets, and Financial Decisions

CHEME 5660 Cornell University

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we answer three questions: How does the course work? Why do financial markets matter even if you never become a trader? What makes one investment preferable to another?

Today's concepts

- Course roadmap: four units move from reading payments to making and testing financial decisions.
- Why markets matter: most adults and Cornell itself are investors, while companies use public markets to raise money.
- Our first decision problem: two investments pay different amounts at different times. How can we compare them?

Note: this lecture has no notebook and no notes packet. These slides are the record.

How This Course Runs

Across twenty-eight lectures, we'll move from understanding financial assets to valuing them, combining them into portfolios, and making investment decisions.

- Materials: download each week from the repository releases page, or clone the repository. After release, files are frozen; corrections are announced.
- Tools: Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. Set these up once, at the start of the semester.
- Questions: ask on the course discussion board rather than by email, so an answer reaches everyone who had the same question.

The course ends with an automated trading project using live data and live market access through the Alpaca Markets application programming interface.

Policies: grading, attendance, academic integrity, and accommodations are posted in full

Markets Connect You, Cornell, and Companies

YOUR SIDE

61%

of U.S. adults had a retirement account in 2025.

Keeping the default mix is still a choice.

That describes today's adults, not your future.

CORNELL'S SIDE

$10.9B

FY2025 endowment

$498M went into FY2025 operations, mainly from endowment payout.

$3.16B of the endowment was designated for financial aid.

COMPANY SIDE

Why go public?

A private company lists shares on a public stock market.

Raise capital: sell new shares; the company receives the money.

Create liquidity: shareholders can trade existing shares.

Sources: Federal Reserve SHED, 2025; Cornell FY2025 Financial Report; Cornell FY2025 Statement of Activities; U.S. SEC

U.S. Markets Move About $1.85 Trillion a Day

To compare annual and daily figures, the daily column divides by 252 trading days. Industry rows show revenue; market rows show cash paid when securities trade.

	Annual total (USD)	Per trading day (USD)
Global biopharma	666B	2.64B
Global batteries	181B	0.72B
Together	847B	3.36B
US debt	304.7T	1.21T
US stocks	153.2T	608B
Prices paid for options	9.3T	37B
US markets together	467.2T	1.85T

Whether measured per year or per trading day, the market total is about 550 times the two-industry total. At that scale, small pricing or risk errors can be costly.

Sources: market sizes 2025 (Fortune Business Insights); turnover 2024–2026 (SIFMA, Cboe)

Gross Notional Value Is About 108 Times Premium Paid

In 2025, buyers paid an average of USD 37B in option premiums each trading day. Those contracts had a much larger gross notional value.

Standard equity options generally represent **100 shares**; index options use a contract multiplier. Across 60.4 million contracts per day, gross notional was about USD 4 trillion:

$$\underbrace{\text{USD 4T}}_{\text{gross notional value}} \Big/ \underbrace{\text{USD 37B}}_{\text{premium paid}} = \underbrace{108\times}_{\text{about \$108 gross notional per \$1 paid}}$$

This ratio approximates exposure-based leverage when $|\delta| \approx 1$, as for a deep in-the-money call. Otherwise, delta determines market exposure and therefore leverage. The USD 4 trillion is gross notional, not cash turnover.

Careful: 108 times is gross notional divided by premium; it approximates leverage only when $|\delta| \approx 1$. About USD 3.0T comes from cash-settled S&P 500 index options; no shares are transferred.

The Course Moves from Valuing Assets to Making Decisions

The first three units teach us to value debt, equity, and derivatives. The fourth uses those tools to build and test financial strategies.

One question connects all four units: what should you do?

Unit 1	Unit 2	Unit 3	Unit 4
Debt	**Equity**	**Derivatives**	**Decisions**
Treasury securities	Stocks and portfolios	Options	Financial strategies
weeks 1–2	weeks 3–7	weeks 8–11	weeks 12–15
compare payments across dates	build portfolios under uncertainty	value investments tied to other values	build and test financial strategies

weeks

This week starts with Treasury securities.

Every Unit Uses the Same Decision Loop

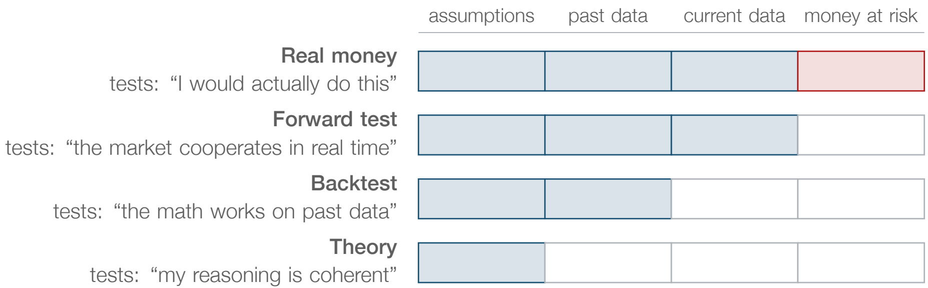
Every unit asks three questions: What do we know now? What can we choose? What happens next? Decision models call these the state, action, and reward.

The course gives us tools to describe each part of the loop and test the links between them.

Careful: every arrow represents an assumption we need to test.

Each Test Exposes a New Way an Idea Can Fail

A model earns trust in stages. Each stage adds evidence and reveals a new way the idea can fail.



A backtest uses past data. A forward test uses current market data, with no money at stake.

Five Investments Have Different Payment Schedules

You can buy each promise for $10,000 today. The payments below use market rates from yesterday. Ignore the names for now; compare only what gets paid and when.

A	$10,028 back, four weeks from today
B	$236 every six months for 10 years, then $10,000
C	$264 every six months for **30** years, then $10,000
D	$271 every six months for 10 years, then $10,000
E	$121 every six months for 10 years, plus an inflation adjustment on every payment and on the $10,000

Which promise would you choose? Vote first; then explain your rule.

As of: published rates for 2026-08-18, each scaled to a $10,000 outlay

Each Payment Schedule Represents a Different Security

	What it is called	Annual rate
A	4-week Treasury bill	3.63%
B	10-year Treasury note	4.71%
C	30-year Treasury bond	5.28%
D	investment-grade corporate bonds	5.42%
E	10-year inflation-indexed Treasury (TIPS)	2.41% real

You compared all five without knowing any of these names. The name identifies the kind of promise; the schedule tells you what gets paid and when. Thursday starts from that distinction.

Careful: D is an index of many investment-grade corporate bonds, not one bond. E's rate is stated after inflation, unlike the others.

The Payment List Leaves Three Questions Open

Choosing the largest listed payment is tempting, but each row hides a different issue.

- C: What if you need the money before year 30? Selling early means accepting the market price then, and that price changes with interest rates. Week 2.
- D: Why does it pay more? Its rate is 5.42%, versus 4.71% for the ten-year Treasury. That gap reflects the chance of missed payments and the difficulty of selling quickly. Units 2 and 3.
- E: What does inflation protection change? Its payments rise with inflation, so its 2.41% rate is not directly comparable with the other rates. Later in the course.

A larger listed payment is not automatically a better investment.

A Bank Can Fail Even When Its Securities Do Not Default

In March 2023 a bank with about $200 billion in assets failed in two days. Its portfolio included one of the five kinds of promises on your list.

Which one, and why?

The security did not default; every payment that came due was made. Yet the bank still failed.

As you watch, identify which promise the bank held and what changed before it failed. Week 2 gives us a model for checking the explanation.

Clip: PBS NewsHour, 28 March 2023, 6 minutes

Summary

Financial decisions start with payment schedules, but the largest payment is not automatically the best choice.

Key takeaways

- **Markets connect investors and companies.** Cornell invests its endowment; going public lets a company raise capital and its shareholders trade.
- **A product name is a label.** What matters first is the payment schedule: what gets paid, when, and under what conditions.
- **Models earn trust one test at a time.** Theory, backtest, forward test, and real money use different evidence and can reveal different failures.

On Thursday, we build our first comparison rule: how to compare money paid at different times.

Further Reading

These readings are optional and will not appear on a problem set. Use them if you want more background on the Federal Reserve, monetary policy, or the Treasury.

- *The Structure and Functions of the Federal Reserve System*, Federal Reserve Board: who does what inside the Federal Reserve.
- *Economy at a Glance: Policy Rate*, Federal Reserve Board: how the Fed reports and carries out interest-rate policy.
- *Fedpoints: Federal Reserve System*, Federal Reserve Bank of New York: a short overview of how the system is organized.
- *Role of the Treasury*, U.S. Department of the Treasury: what the Treasury does and how it differs from the Federal Reserve.